On the Residual Bias of Subclassification for Continuous Confounders: A Causal Perspective

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Abstract

In observational studies, controlling for continuous confounders through discretization (subclassification or stratification) is a common empirical strategy. This paper provides a formal mathematical proof that approximating a continuous hidden confounder via stratified regression only partially reduces confounding bias. We derive the exact expression for the residual "within-bin" omitted variable bias and demonstrate that averaging local regressions does not yield a strictly unbiased estimate for a finite number of strata. Finally, we map this methodological approach to Judea Pearl's Backdoor Adjustment Formula, illustrating that the residual bias is mathematically equivalent to the error term in a Riemann sum approximation of the continuous backdoor integral.

1 Introduction

A foundational technique in causal inference is subclassification (or stratification), which attempts to block backdoor paths by grouping a confounder into discrete bins and estimating effects locally within those bins. While this intuition is highly practical, to be mathematically precise, averaging these local regressions will not yield a strictly unbiased estimate for a finite number of bins (N). It reduces the bias significantly, but a residual bias remains unless the number of bins approaches infinity ($N \to \infty$) or the true confounder is perfectly discrete and perfectly captured by the bins (1). This paper formally derives why this bias shrinks but does not disappear completely.

2 The Model Setup

Let the true data-generating process be defined by the following linear model:

$$y_i = \beta x_i + \gamma z_i + \epsilon_i \tag{1}$$

where we wish to estimate the causal parameter β. We make the following assumptions:

- z_i is an unobserved continuous confounder.
- x_i and z_i are correlated, meaning $\mathrm{Cov}(x, z) \neq 0$.
- x_i is exogenous to the error term, $\mathrm{Cov}(x, \epsilon) = 0$.
- We only observe or estimate a discretized version of z, denoted as z', which is grouped into N distinct bins: $z' \in \{1, 2, \ldots, N\}$.

For example we might know which rows of the data could be more exposed to the confounder z.

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3 Deriving the Local Estimator and Within-Bin Bias

Suppose we run N local regressions of y on x for each bin. Considering a single bin, $z' = k$, the local Ordinary Least Squares (OLS) estimator for the effect of x on y inside this specific bin is $\hat{\beta}_k$. The population equivalent (probability limit) of this estimator is given by the ratio of the conditional covariance to the conditional variance:

$$\beta_k = \frac{\text{Cov}(x, y \mid z' = k)}{\text{Var}(x \mid z' = k)} \tag{2}$$

Substituting the true model $y_i = \beta x_i + \gamma z_i + \epsilon_i$ into the numerator yields:

$$\beta_k = \frac{\text{Cov}(x, \beta x + \gamma z + \epsilon \mid z' = k)}{\text{Var}(x \mid z' = k)} \tag{3}$$

Using the properties of covariance, this expands to:

$$\beta_k = \beta \frac{\text{Cov}(x, x \mid z' = k)}{\text{Var}(x \mid z' = k)} + \gamma \frac{\text{Cov}(x, z \mid z' = k)}{\text{Var}(x \mid z' = k)} + \frac{\text{Cov}(x, \epsilon \mid z' = k)}{\text{Var}(x \mid z' = k)} \tag{4}$$

Since $\text{Cov}(x, x) = \text{Var}(x)$ and $\text{Cov}(x, \epsilon) = 0$, the expression simplifies to:

$$\beta_k = \beta + \gamma \frac{\text{Cov}(x, z \mid z' = k)}{\text{Var}(x \mid z' = k)} \tag{5}$$

To obtain an unbiased estimate of β in this local bin, the second term must equal exactly zero. However, $\text{Cov}(x, z \mid z' = k) \neq 0$. Because z is continuous, it continues to vary inside the bounds of the bin $z' = k$. Since x and z are correlated globally, they will almost certainly remain correlated within the narrow slice of the bin. This term represents the *within-bin omitted variable bias.*

When averaging the N local estimators to compute an overall estimate $\hat{\beta}_{avg}$, one calculates a weighted average of these local β_k values:

$$\hat{\beta}_{avg} = \sum_{k=1}^{N} W_k \beta_k = \beta + \gamma \sum_{k=1}^{N} W_k \frac{\text{Cov}(x, z \mid z' = k)}{\text{Var}(x \mid z' = k)} \tag{6}$$

where W_k represents the weighting scheme (often proportional to the variance of x in bin k or the sample size of bin k). Because the within-bin covariances are non-zero, the sum of these biases is also non-zero. Therefore, the averaged estimate remains strictly biased.

However we get a less biased estimate than if we had just regressed y on x for all rows of the data.

4 Conditions for Unbiasedness

The subclassification method successfully extracts an unbiased estimate of x only under specific, limiting conditions:

- **Asymptotic Bins ($N \to \infty$):** If the bins become infinitely small, the variance of z within any given bin shrinks to exactly zero. If z is constant within the bin, $\text{Cov}(x, z \mid z' = k) = 0$, and the bias disappears. This mathematical reality underscores why controlling for a continuous variable using flexible parametric functions (e.g., polynomials or splines) is generally preferred over binning.

- **The Confounder is Actually Discrete:** If the true z only ever took on N values (e.g., categorical data), and z' perfectly mapped those values, there is no within-bin variation of z, yielding zero bias.

- **No Conditional Correlation:** If x and z happen to be independent within every bin, despite being dependent globally.

In summary, discretizing a continuous confounder into N bins and averaging local regressions successfully removes the between-bin confounding, massively reducing overall error. However, it leaves behind within-bin confounding, maintaining strict bias for any finite N.

5 Connection to Pearl's Backdoor Adjustment Formula

The empirical strategy of running local regressions across bins of a confounder and averaging the results is the direct empirical application of Judea Pearl's Backdoor Adjustment Formula (2). According to Pearl's causal framework, if a set of variables Z satisfies the backdoor criterion relative to a treatment X and an outcome Y, the causal effect of intervening on X (*do*-operator) is calculated by marginalizing the conditional distribution of Y over the prior distribution of the confounder Z.

For a discrete confounder Z:

$$P(Y = y \mid do(X = x)) = \sum_z P(Y = y \mid X = x, Z = z)P(Z = z) \tag{7}$$

For a continuous confounder Z:

$$P(Y = y \mid do(X = x)) = \int P(Y = y \mid X = x, Z = z)P(Z = z)dz \tag{8}$$

The components of the subclassification method align perfectly with this formula:

- $P(Y = y \mid X = x, Z = z)$ maps to the local regression. Isolating data inside a specific bin ($z' = k$) estimates the conditional relationship between x and y while holding the confounder constant.
- $P(Z = z)$ represents the weighting of the bins. The weight W_k applied to bin k acts as the empirical estimate of the marginal probability of Z.
- The final averaging step ($\sum_z$) executes the marginalization required to extract the global Average Treatment Effect.

5.1 Why the Bias Persists: A Riemann Sum Perspective

Viewing this through the Backdoor Formula provides an intuitive understanding of the residual bias. Because Z is a continuous variable, identifying the true causal effect requires the integral form of Pearl's formula. By discretizing Z into N bins (Z'), the subclassification methodology forcibly applies the summation form of the formula:

$$P(Y \mid do(X)) \approx \sum_{k=1}^{N} P(Y \mid X, z'_k)P(z'_k) \tag{9}$$

Substituting a continuous integral with a discrete summation over N bins effectively performs a Riemann sum approximation. The lost mathematical precision between the exact integral and the discrete summation is exactly the within-bin omitted variable bias derived in Section 3. As $N \to \infty$, the discrete summation converges into the continuous integral, and the bias drops to zero.

6 Conclusion

We have demonstrated that while subclassification is a powerful empirical tool for mitigating omitted variable bias, it is insufficient for achieving complete unbiasedness when applied to continuous confounders. The residual within-bin variation acts as an unobserved confounder at the local level. Recognizing this limitation is crucial for applied researchers, particularly when framing estimation strategies within non-parametric frameworks like Pearl's do-calculus.

References

[1] Cochran, W. G. (1968). The effectiveness of adjustment by subclassification in removing bias in observational studies. *Biometrics*, 24(2), 295–313.

[2] Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.